

Name: _____

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Homework #02: Registers and Directives

Directives

Use the code below as a reference for the problem 1-5

```
1  ;***** Program Setup Code *****
2  THUMB
3
4  AREA      MyConstants, DATA, READONLY, ALIGN=1      ;
5
6  ALIGN
7
8  AREA      MyVariables, DATA, READWRITE, ALIGN=2      ;
9
10 ALIGN
11
12 AREA |.text|, CODE, READONLY, ALIGN=3      ;
13
14 EXPORT Start
...

```

(5 pts each)

1. Which directive instructs the assembler to interpret subsequent instructions and Thumb or Thumb-2 instructions.

ANS: _____

2. Constants created in the MyConstants area would typically reside in _____ since their values cannot change during program execution.

- a. RAM
- b. ROM

3. In the code above, line 8 creates a DATA area in _____.

- a. RAM

- b. ROM
4. In the code above, the `ALIGN=3` directive on line 12 ensures that the area is reserved in memory is aligned on a(n) _____.
 - a. 1-byte boundary
 - b. 2-byte boundary
 - c. 4-byte boundary
 - d. 8-byte boundary
 5. In the code above, the `ALIGN` directive on line 10 ensures that the data or instructions that follows is assigned on a(n) _____.
 - a. 1-byte boundary
 - b. 2-byte boundary
 - c. 4-byte boundary
 - d. 8-byte boundary
 6. Write the directive line to reserve an area in memory named `myStorage`. The area should start at an address that is a multiple of 8 and will store variables used in your program.

General Register knowledge

7. How many general-purpose registers does the Cortex-M4 (the microprocessor in our microcontroller chip) have in its register bank? (do not include the special purpose register `SP`, `LP`, & `PC` in your answer).

ANS: _____

8. How many **bits** are in a general-purpose register on the Cortex-M4 microprocessor?

ANS: _____

Loading registers (move immediate numbers into registers)

9. Write the single assembly instruction to load register **R2** with the immediate binary value `2_10011010`.

ANS: _____

10. Write the single assembly instructions to load register **R4** with the immediate decimal value 14.

ANS: _____

11. Write the single assembly instruction to load register **R7** with immediate hex value 0xAB23CD36.

ANS: _____

12. Write the single assembly instruction to load register **R5** with the value currently stored in register **R2**.

ANS: _____

Arithmetic operations

13. Write the single instruction to add the immediate hexadecimal value 0x32F to the value in register **R3** and store the result in to register **R0**.

ANS: _____

14. Write the single instruction to subtract the immediate decimal value 12 from the values currently in **R2** and store the result in register **R5**.

ANS: _____

15. Write the single instruction to add the values stored in register **R1** and register **R3** and store the result in to register **R0**.

ANS: _____

16. Write the single instruction to subtract the values currently in register **R1** from the values currently in **R3** and store the result in register **R3**.

ANS: _____

17. Write the single instruction to multiply the value currently in register **R5** by the value currently in register **R6** and store the result in register **R4**.

ANS: _____

18. Write the single instruction to divide the value currently in register **R6** by the value currently in register **R7** and store the result in register **R8**. (unsigned division).

ANS: _____

19. (10 pts) Write the ARM assembly code to perform the following C-like instruction. You are only allowed to use register 0 and 1 in your code. (R0 is actually register 0 in this case and not a variable in RAM.)

$R0 = 7 + 3 * 3$